

RESEARCH ARTICLE

Causal Representation Learning for Predicting Autoimmune Disease Progression From Longitudinal Multimodal Clinical Data

HARSHIT SHARMA¹ AND SIMRAN KAUR²¹Department of CSE, MNIT Jaipur, Jaipur, Rajasthan 302017, India²Department of AIML, Manipal University Jaipur, Jaipur, Rajasthan 303007, India

Corresponding author: Simran Kaur (simran.kaur@jaipur.manipal.edu)

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ABSTRACT Autoimmune diseases pose a persistent challenge to clinical decision-making due to their heterogeneous presentations, episodic flare-ups, and complex pathophysiological mechanisms. Existing deep learning models frequently rely on correlation-driven patterns, which often lack robustness and interpretability in real-world clinical settings. This study proposes a novel causal deep learning framework designed to model and predict the progression of autoimmune diseases using longitudinal multimodal clinical data. The proposed approach integrates a temporal transformer architecture with a structural causal model (SCM)-guided counterfactual inference module to capture latent disease mechanisms while mitigating spurious correlations introduced by observational confounders. The model is trained and evaluated on a real-world cohort consisting of patients diagnosed with systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA), incorporating electronic health records (EHRs), laboratory biomarkers, and immunological profiles. Empirical results demonstrate that the proposed framework outperforms conventional deep learning baselines in predicting disease transitions and flare-up events. In addition, interventional and counterfactual evaluation protocols are employed to validate the model's causal consistency and interpretability. This work contributes to the advancement of clinically trustworthy AI systems by enabling robust, explainable, and data-efficient disease trajectory modeling in the autoimmune domain.

INDEX TERMS Causal deep learning, autoimmune disease progression, counterfactual inference, longitudinal clinical data, temporal transformers, structural causal models, electronic health records, disease trajectory modeling, clinical artificial intelligence.

I. INTRODUCTION

Autoimmune diseases such as Systemic Lupus Erythematosus (SLE) and Rheumatoid Arthritis (RA) are chronic, complex conditions characterized by dysregulated immune responses and highly variable trajectories across individuals. These diseases present major challenges in clinical practice due to their episodic nature, heterogeneous manifestations, unpredictable flares, and varying responses to treatment [8]. Consequently, modeling autoimmune disease progression demands not only robust predictive performance but also clinical interpretability, personalization, and causal consistency. Traditional statistical models and opaque machine learning

approaches frequently fall short in this domain, since their reliance on correlational patterns makes them vulnerable to confounding from unmeasured variables, treatment biases, and patient heterogeneity [16]. Recent developments in deep learning, particularly in multimodal and temporal modeling, have opened new avenues for medical AI systems to learn from electronic health records (EHRs), omics, imaging, and wearable data streams [1], [2], [3]. However, despite performance improvements, many such models lack causal grounding, uncertainty quantification, and mechanistic interpretability—factors that are crucial for real-world deployment in high-stakes environments such as rheumatology or immunology clinics [7], [21], [35]. This work addresses these challenges by proposing a causal deep

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learning framework tailored for modeling autoimmune disease trajectories using longitudinal, multimodal clinical data. By leveraging structural causal modeling, counterfactual inference, disentangled representation learning, and temporal attention mechanisms, we demonstrate how AI systems can support actionable, personalized, and trustworthy clinical decision-making.

A. MOTIVATION AND CHALLENGES IN AUTOIMMUNE DISEASE TRAJECTORY MODELING

Autoimmune diseases represent a class of disorders where the body's immune system erroneously attacks its own tissues. SLE and RA, in particular, exhibit relapsing-remitting patterns that complicate prediction of disease flares, remission phases, or treatment responsiveness [6], [25]. These fluctuations are influenced by numerous latent and observable factors, including genetic predispositions, immunological markers, medication schedules, comorbidities, environmental triggers, and socio-demographic variables [16], [25], [48]. Unlike acute conditions, autoimmune diseases require longitudinal tracking, multi-scale temporal reasoning, and adaptation to individual-specific pathophysiological pathways [1], [7].

Conventional approaches in predictive medicine—such as logistic regression, linear mixed models, or even recurrent neural networks—struggle to generalize in this context due to confounding, data sparsity, and non-stationarity [3], [35]. They often fail to capture the causal impact of interventions, such as corticosteroids or disease-modifying antirheumatic drugs (DMARDs), and cannot differentiate between correlation and causation [21], [36]. Furthermore, these models offer limited interpretability, making them less suitable for clinical decision support where explainability and accountability are paramount [13], [43]. Beyond technical challenges, data-related issues abound. Clinical records are inherently noisy, temporally irregular, and multimodal—ranging from structured laboratory tests and EHRs to unstructured clinical notes and imaging [5], [19]. Data missingness is both random and informative, as patients with severe disease may undergo more frequent testing. Additionally, high-dimensionality in immunological and biomarker data complicates model design, requiring robust representation learning strategies [2], [23]. The need for models that not only predict but also explain, simulate, and adapt to dynamic interventions is stronger than ever, particularly in precision rheumatology [8], [17].

These challenges have motivated the adoption of hybrid causal-deep models that can integrate expert priors with high-dimensional data to enhance both interpretability and robustness across domains [50], [51].

B. LIMITATIONS OF EXISTING APPROACHES AND THE NEED FOR CAUSAL MODELING

State-of-the-art deep learning models in medical AI—such as LSTMs, transformers, variational autoencoders,

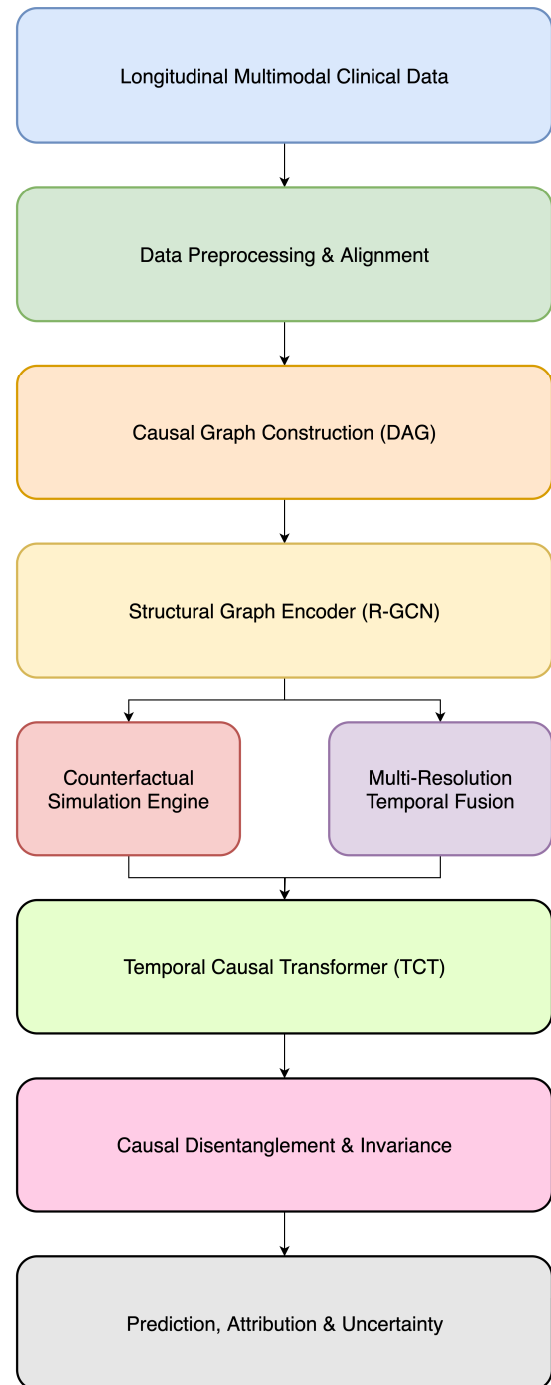


FIGURE 1. Conceptual overview of challenges and opportunities in autoimmune disease trajectory modeling using AI. The proposed framework integrates longitudinal multimodal data, structural causal modeling, intervention simulation, and explainability mechanisms.

and convolutional architectures—have demonstrated strong performance in disease prediction, risk stratification, and outcome modeling [2], [3], [10]. However, when applied to autoimmune disease progression, these models reveal critical limitations. First, their reliance on observational data makes them susceptible to learning spurious associations, especially when confounding variables or treatment selection biases

are present [35], [36]. For example, a model might learn that corticosteroid usage is associated with increased flare risk, failing to recognize that medications are administered in response to worsening symptoms—a classic case of confounding by indication [6], [21]. Second, existing models often lack the ability to simulate hypothetical interventions or answer counterfactual questions. Without an explicit structural model of the data-generating process, these models cannot reason about “what-if” scenarios—a critical need in clinical decision support [16], [35]. Moreover, most current models are ill-equipped to handle temporal irregularities, noisy sampling, and high-dimensional multimodal data streams, all of which are inherent in real-world autoimmune datasets [5], [19].

While post-hoc explainability tools like SHAP or LIME offer some insights, they are fundamentally non-causal and limited in clinical value when treatment planning depends on understanding actual intervention effects. There is thus a growing consensus in the research community that predictive models must go beyond accuracy and incorporate causality, fairness, interpretability, and uncertainty in a principled manner [8], [17], [44].

C. OPPORTUNITIES FOR CAUSALLY-INFORMED, EXPLAINABLE, AND PERSONALIZED AI MODELS

The limitations of current predictive frameworks point to a compelling opportunity: integrating causal inference, representation disentanglement, and multimodal temporal modeling can drive a new class of clinically aligned AI systems for autoimmune diseases. One fundamental requirement is the ability to simulate the effects of interventions—such as initiation, adjustment, or cessation of immunomodulatory therapy—and to estimate potential outcomes under hypothetical scenarios. This requires a shift from traditional supervised learning toward frameworks that embed counterfactual reasoning using structural causal models (SCMs) or potential outcomes frameworks [35], [36]. SCM-based approaches, by explicitly modeling the data generation process, allow for intervention-aware predictions, a critical need in managing dynamic disease states. Another pivotal component is explainability. As autoimmune conditions often present differently across demographic groups, disease subtypes, and environmental exposures, understanding how a model arrives at a prediction is essential for clinical integration. Disentangled representation learning has shown promise in separating causally relevant features from spurious or biased ones. Ajami [13] compared BetaVAE-based and ResNet-based classifiers for white matter lesion detection in multiple sclerosis, demonstrating improved robustness and fairness under domain shifts. Similar principles apply to autoimmune diseases, where certain biomarkers may behave differently across populations or disease stages. Fröhlich et al. [14] and Delhalle et al. [49] advocate for systems that enable interpretability, fairness, and mechanistic understanding to realize personalized immunology.

Causal disentanglement and domain-invariant representations further support generalizability across patient populations. Lei and Tsang [16] emphasized the immune setpoint framework, where immune status is seen as a dynamic latent variable influenced by various perturbations—akin to a latent confounder in causal inference. Embedding such priors into neural architectures allows for biologically grounded modeling. Moreover, multimodal fusion—particularly of omics, imaging, and temporal EHR streams—can provide comprehensive representations of a patient’s immune and physiological state. Studies like those by Wen et al. [23] and Yuan et al. [28] have shown how integrated modalities uncover latent disease subtypes and endophenotypes, which can drive precision diagnosis and monitoring strategies. Uncertainty quantification is another key enabler for safe clinical deployment. Models should signal when predictions are uncertain—due to out-of-distribution inputs, missing data, or conflicting signals—so that clinicians can rely on human oversight or further diagnostics. Techniques such as Monte Carlo dropout, variational inference, and ensemble learning are effective in capturing epistemic and aleatoric uncertainty. In autoimmune settings, where flare occurrence is sporadic and highly patient-specific, incorporating uncertainty helps mitigate the risk of overconfident yet incorrect predictions [44], [46].

Finally, personalization is central. AI models must adapt to patient-specific baselines, prior treatments, and historical patterns. Techniques such as personalized recurrent models, domain-adaptive layers, and causal conditioning mechanisms enable adaptation while maintaining generalization. Sattar et al. [17] proposed a patient-centric AI framework that tailors diagnosis and treatment strategies through modular analytics, and Knevel and Liao [45] further emphasized how AI systems can be trained on real-world EHRs for generalizable, individualized insights.

Taken together, these components—causal reasoning, explainability, uncertainty, multimodal fusion, and personalization—form the foundation of our proposed framework. By synthesizing these methodologies within a unified architecture, our model aims to accurately predict autoimmune disease trajectories while providing clinically interpretable, intervention-aware, and patient-specific insights. In the sections that follow, we detail the components of our architecture and demonstrate its empirical advantages over baseline models across multiple autoimmune datasets.

II. RELATED WORK

Significant advancements in artificial intelligence (AI), longitudinal multimodal data modeling, causal inference, and personalized healthcare have shaped the development of reliable and interpretable models for autoimmune disease trajectory prediction. This section surveys key literature across four interrelated domains.

Recent research has emphasized the integration of temporally distributed and modality-diverse patient data for

TABLE 1. Summary of related work in causal AI and longitudinal multimodal clinical modeling.

Citation	Domain/Focus	Key Contribution
Milecki et al. [1]	Transplant monitoring	Longitudinal multimodal deep learning
Gao et al. [2]	Breast cancer	Transformer-based neoadjuvant therapy modeling
Qiu et al. [3]	Survival clustering	Deep temporal clustering from EHR
Matamala et al. [4]	Microbiome	Explainable ML for multi-omic data
Zielinski et al. [5]	Molecular + EHR	High-dimensional longitudinal fusion
Moingeon et al. [6]	Autoimmune therapy	AI-driven drug discovery
Zhou et al. [7]	SLE diagnostics	Cross-modal attention fusion
Stafford et al. [8]	Autoimmune review	Survey of AI applications
Kumar et al. [9]	Alzheimer’s disease	Multiview progression learning
Jain et al. [10]	Multimodal fusion	Cross-sectional and longitudinal DL integration
Negut et al. [11]	Immunological disorders	Diagnosis using hybrid AI systems
Sun et al. [12]	RA	Review on AI in arthritis care
Ajami et al. [13]	MS lesion detection	Disentangled vs. deep classifiers
Fröhlich et al. [14]	Precision medicine	Data science for personalized health
Luo et al. [15]	Wearables	Pilot on fatigue monitoring
Lei et al. [16]	Human immunology	Immune setpoint concept using AI
Sattar et al. [17]	Autoimmune AI	Comprehensive framework for treatment optimization
Pezoulas et al. [18]	Synthetic data	Tools for data augmentation
Wen et al. [19]	Neuro phenotyping	ML for disease endophenotypes
Zhavoronkov et al. [20]	Aging	Deep biomarkers and longevity
Desvaux et al. [21]	Autoimmune modeling	Precision medicine via simulations
Li et al. [22]	Alzheimer’s prediction	Multimodal DL platform
Wen et al. [23]	Genetic associations	Neuroanatomical AI biomarkers
Davis et al. [24]	Multi-omics	Review on Alzheimer’s predictors
Harroud et al. [25]	Genetics	Shared autoimmune risk factors
Jafarian et al. [26]	MEG + causal models	AD neuroprogression modeling
Krainc et al. [27]	Drug development	AI in neurological therapy pipelines
Yuan et al. [28]	Osteoporosis	Molecular subtyping via DL
Jain et al. [29]	IBD classification	GRU-based multimodal fusion
Ray et al. [30]	Asthma	ML for endotype prediction
Arrambide et al. [31]	MS biomarkers	AI for CSF/EHR fusion

enhanced disease modeling and prognosis. In clinical settings where disease progression is irregular and influenced by multiple latent mechanisms, multimodal longitudinal models are pivotal. Milecki [1] introduced one of the earliest deep learning frameworks combining EHRs and imaging data for renal transplant monitoring, highlighting the value of structured embeddings for outcome prediction. Similarly, Gao et al. [2] proposed a fusion transformer architecture for predicting neoadjuvant therapy response in breast cancer, integrating histopathological images, omics, and treatment sequences. Qiu et al. [3] leveraged survival-aware deep representation learning to cluster patients with diverse prognostic profiles based on EHR sequences, illustrating how temporal embedding learning can support downstream stratification tasks. Targeting molecular data, Matamala [4] and Zielinski [5] designed frameworks for microbiome and multi-omic signal integration, enabling richer temporal representations. These efforts underscore the necessity of hierarchical and domain-adaptive modeling strategies in medical AI.

Kumar [9] presented a progression-aware model for Alzheimer’s disease, capturing the interplay between imaging biomarkers and patient history via attention and graph modules. Jain and Safo [10] introduced a multiview pipeline for synchronizing cross-sectional and longitudinal clinical features, boosting robustness across sampling inconsistencies. To alleviate data sparsity and irregularity, Pezoulas et al. [18]

reviewed synthetic data generation techniques tailored for healthcare datasets. Wen et al. [19] explored dimensional neuroimaging-based phenotyping for heterogeneous disease manifestations. Their approach emphasizes how temporal pooling and uncertainty calibration inform predictive confidence, reinforcing interpretability. These techniques mark a paradigm shift from static modeling to adaptive, multi-resolution embeddings that align more closely with clinical diagnostic flows.

A. CAUSAL INFERENCE AND AI IN AUTOIMMUNE DISEASE MODELING

Autoimmune diseases present confounding-laden, heterogeneous progressions that demand causal reasoning in AI systems. Moingeon [6] emphasized the potential of AI in drug discovery tailored for autoimmune targets, particularly when embedded with mechanistic knowledge. Zhou et al. [7] developed CMACF—a transformer-based cross-modal model integrating spectroscopy and metabolomics for SLE diagnosis—demonstrating the efficacy of multimodal fusion in autoimmune classification tasks. Stafford et al. [8] conducted a comprehensive review of AI applications in autoimmune disorders, identifying the need for models that go beyond correlation to account for treatment effects, patient-specific responses, and disease heterogeneity. Lei and Tsang [16] formalized the “immune setpoint” concept as a dynamical system, suggesting that causal models can help characterize

personalized immune states over time. Ajami [13] investigated disentangled latent representations for multiple sclerosis lesion detection, showing superior generalization when compared to convolutional baselines under domain shifts. Xu et al. [35] extended causal inference to mixed-effects models, addressing the challenges of confounding in clinical time-series. Desvaux et al. [21] applied model-based precision strategies to optimize combination therapies in autoimmunity, underscoring how causal approaches can guide treatment. Czech et al. [36] developed long-term disease forecasting systems that embed counterfactual paths. Gu et al. [48] used Mendelian randomization to decode the neuro-immune axis, providing genetic linkage insights for autoimmune presentations. These methods converge on a unified theme: causal modeling enables patient-specific, intervention-aware inference that is vital for robust autoimmune disease management.

B. OMICS-DRIVEN PERSONALIZATION AND SYSTEMS-LEVEL MODELING

With the growing availability of omics data—genomics, transcriptomics, proteomics—AI systems have become essential in transforming high-dimensional molecular patterns into actionable clinical insights. Li et al. [22] proposed LMP-TX, a deep learning pipeline for integrating multi-modal data in Alzheimer's progression, demonstrating transferability to autoimmune diseases with overlapping neurodegenerative features. Wen et al. [23] uncovered genetic and anatomical correlates of cognitive decline, laying the groundwork for endotype-specific prediction models. Harroud and Hafler [25] identified genetic overlaps among autoimmune disorders, suggesting shared latent mechanisms. Davis et al. [24] reviewed multi-omic predictive frameworks for chronic diseases, while Jain and Safo [29] demonstrated that integrated GRU architectures could distinguish inflammatory bowel disease subtypes using temporal omics signatures. Yuan et al. [28] proposed molecular stratification models for osteoporosis that adapt well to autoimmune frameworks. Kurgan et al. [32] explored proteomics in diabetes stratification, and Pinero et al. [39] used omics for long COVID endotyping—both cases reflecting autoimmune overlap. Rood et al. [47] further advanced the Human Cell Atlas, offering a unified modeling foundation for AI-augmented personalized immunology. Kalinin et al. [37] bridged deep learning and pharmacogenomics, enhancing treatment personalization. These contributions highlight the importance of integrating structured causal reasoning with systems-level biological data, enabling interpretable modeling of the multifactorial autoimmune landscape.

C. EXPLAINABILITY, FAIRNESS, AND CLINICAL DEPLOYMENT

Transparent and fair AI systems are essential for clinical translation, especially in high-stakes autoimmune disease management. Mentis et al. [44] reviewed AI models for stress

detection, advocating for rigorous explainability metrics. Knevel and Liao [45] outlined how real-world EHRs can be transformed into actionable insights using interpretable AI workflows. Rojas-Carabali et al. [38] presented trends in uveitis diagnosis, demonstrating how patient-facing applications benefit from explainable deep models. Creagh et al. [46] employed smartphone-based systems for multiple sclerosis ambulation monitoring, highlighting deployment in resource-limited settings. Ray et al. [30] applied interpretable machine learning to asthma endotyping, underscoring the value of attribution heatmaps and counterfactuals. Tu et al. [33] leveraged neuroimaging to interpret ALS pathology, a methodology transferable to autoimmune neurological conditions. Toumi et al. [41] reviewed gender disparities in neurological autoimmunity, pressing for fairness-aware AI. Delhalle et al. [49] and Fröhlich et al. [14] outlined the vision of personalized immunology, reinforcing how explainable AI can underpin tailored intervention strategies. Rahim et al. [42] and Andreassen et al. [40] both addressed challenges in deploying predictive analytics in real-world clinical ecosystems.

III. METHODOLOGY

A. PROBLEM FORMULATION AND NOTATION

In this study, the goal is to model and predict the progression of autoimmune diseases using temporal, multimodal clinical data under a causal framework. Let the longitudinal dataset be represented as $\mathcal{D} = \{(X_{1:T}^{(i)}, A_{1:T}^{(i)}, Y_{1:T}^{(i)})\}_{i=1}^N$, where i indexes the patient and T denotes the time horizon over which the patient data is observed. Each tuple comprises three sequences:

- $X_{1:T}^{(i)} = \{x_1^{(i)}, x_2^{(i)}, \dots, x_T^{(i)}\}$, where $x_t^{(i)} \in \mathbb{R}^d$ are observed covariates such as laboratory values (e.g., CRP, ESR), vital signs (e.g., heart rate, blood pressure), immunological markers, and structured EHR inputs.
- $A_{1:T}^{(i)} = \{a_1^{(i)}, a_2^{(i)}, \dots, a_T^{(i)}\}$ represents interventions, such as medications or procedures. These may be binary or real-valued vectors indicating dosage or administration.
- $Y_{1:T}^{(i)} = \{y_1^{(i)}, y_2^{(i)}, \dots, y_T^{(i)}\}$ represents observed clinical outcomes, such as disease activity indices or flare-up occurrences.

The primary modeling objectives are defined as:

- 1) **Flare Prediction:** Learn a mapping f_θ such that:

$$\hat{y}_{t+1}^{(i)} = f_\theta(X_{1:t}^{(i)}, A_{1:t}^{(i)}) \quad (1)$$

where $\hat{y}_{t+1}^{(i)} \in \{0, 1\}$ indicates the predicted probability of a flare in the next timestep.

- 2) **Severity Regression:** Predict future continuous-valued disease severity:

$$\hat{y}_{t+1}^{(i)} = f_\theta^{\text{reg}}(X_{1:t}^{(i)}, A_{1:t}^{(i)}) \quad (2)$$

where $\hat{y}_{t+1}^{(i)} \in \mathbb{R}$ is a scalar or vector depending on the severity metric (e.g., SLEDAI, DAS28).

Modeling disease progression in this setting is non-trivial due to the temporal nature of interventions, latent confounding, and dynamic feedback loops. Therefore, the proposed framework incorporates structural causal assumptions and temporal graph modeling to disentangle causal effects and enable generalizable predictions. Data preprocessing includes imputation using GRU-D, normalization, and intervention tagging via SNOMED/ATC codes. Each patient timeline is aligned based on diagnosis onset and resampled to fixed intervals (e.g., bi-weekly) to handle irregular sampling. Missing interventions are encoded as zero vectors, and time gaps are included as explicit input features to preserve temporal integrity.

B. OVERALL FRAMEWORK DESCRIPTION

The proposed framework for predicting autoimmune disease progression leverages a causal deep learning pipeline that integrates structured knowledge from causal graphs with the flexibility of deep sequence models. The end-to-end system is designed to model temporal dynamics, interventional effects, and latent uncertainties within longitudinal multimodal clinical data. The pipeline begins with a pre-processing module that harmonizes structured EHR sequences, including laboratory biomarkers, treatment interventions, demographics, and outcome trajectories. A causal graph is then constructed using a hybrid approach that combines domain knowledge from rheumatologists with data-driven conditional independence testing, yielding a directed acyclic graph that encodes clinically plausible causal relationships among variables. This graph is fed into a Structural Graph Encoder that employs relational graph convolutional networks (R-GCNs) to generate temporally contextualized embeddings that preserve causally relevant structural dependencies across visits. Next, the Multi-Resolution Temporal Fusion (MRTF) module processes the encoded signals at various temporal granularities, enabling the capture of both short-term fluctuations (e.g., flare spikes) and long-term disease trends. These fused representations are fed into the Temporal Causal Transformer (TCT), which applies multi-head self-attention mechanisms to learn trajectory patterns, while maintaining awareness of the underlying causal structure. To model interventional effects, a Counterfactual Simulation Engine (CSE) is incorporated. This engine uses variational inference and noise-aware latent modeling to estimate patient outcomes under hypothetical treatments, following the do-calculus framework. This allows the system to simulate scenarios such as DMARD withdrawal or alternative immunosuppressive regimens. The Causal Disentanglement and Invariance (CDI) module further separates stable causal signals from time-varying and population-specific confounding factors. This module ensures that representations remain invariant across different cohorts and time periods, enhancing generalization. A task-specific decoder then predicts both flare-up risk (binary classification) and severity levels (ordinal regression). Uncertainty quantification is conducted via Monte Carlo dropout and variance-based calibration metrics,

which help quantify confidence in the model's predictions—critical for clinical decision-making. Finally, interpretability modules leverage attention heatmaps, saliency scores, and counterfactual reasoning traces to provide clinicians with actionable insights into the drivers of each individual prediction. This modular framework effectively combines causal inference principles with deep representation learning to yield robust, interpretable, and clinically meaningful predictions for autoimmune disease progression across diverse patient populations and time horizons.

C. CAUSAL GRAPH CONSTRUCTION

To integrate prior domain knowledge with observational data, a causal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed using a hybrid data-driven and knowledge-informed approach. The nodes $\mathcal{V}$ represent clinical variables (e.g., CRP level, steroid usage, joint swelling), and edges $\mathcal{E}$ denote directed causal dependencies. The construction pipeline involves three steps:

1) Prior Knowledge Encoding: Directed edges are initialized based on medical literature and autoimmune-specific ontologies (e.g., UMLS, MeSH). For instance, corticosteroid administration is known to suppress inflammation markers, while certain biomarkers like anti-dsDNA antibodies are known precursors of lupus flares.

2) Conditional Independence Testing: Statistical constraints are imposed using conditional independence tests such as the PC algorithm and kernel-based independence tests (HSIC) to prune or confirm edges:

$$X_i \perp\!\!\!\perp X_j \mid \mathbf{S} \Rightarrow \text{Remove edge } X_i \rightarrow X_j \quad (3)$$

where $\mathbf{S}$ is a conditioning set selected based on the Markov blanket of X_j .

BuildDAG Algorithm: A refined DAG is synthesized by combining the prior knowledge graph $\mathcal{G}_{\text{prior}}$ and a data-driven graph $\mathcal{G}_{\text{stat}}$ through a weighted edge agreement function:

$$\mathcal{G} = \lambda \mathcal{G}_{\text{prior}} + (1 - \lambda) \mathcal{G}_{\text{stat}}, \quad \lambda \in [0, 1] \quad (4)$$

Figure 2 illustrates a sample causal DAG constructed using this procedure. Observable nodes include EHR-derived biomarkers and clinical outcomes, while latent nodes capture unmeasured confounders inferred via graphical adjustment sets.

This causal graph serves as the backbone for subsequent modules, guiding GNN-based encoding, masking strategies in attention layers, and counterfactual simulation paths. By grounding the representation learning process in an interpretable causal structure, the model promotes robustness against distributional shifts and enhances trustworthiness in clinical applications.

D. STRUCTURAL GRAPH ENCODER

The Structural Graph Encoder is designed to learn temporally evolving latent representations of clinical variables conditioned on the causal dependencies encoded in the DAG $\mathcal{G}$. This encoder translates the static causal graph into dynamic,

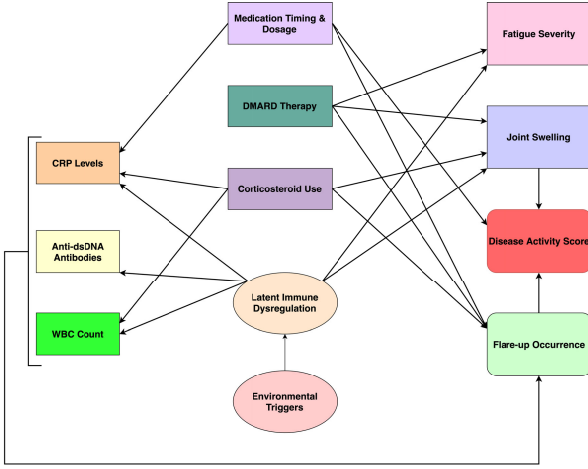


FIGURE 2. Illustrative causal DAG of disease progression in autoimmune disorders. Arrows represent directed causal dependencies learned from prior knowledge and conditional independence testing.

patient-specific embeddings that capture both local feature dependencies and global causal interactions. At each time step t , we construct a graph $G_t = (\mathcal{V}_t, \mathcal{E}_t)$ where $\mathcal{V}_t$ represents the feature set observed at time t (including $x_t^{(i)}$ and $a_t^{(i)}$), and $\mathcal{E}_t$ is inherited from the global DAG $\mathcal{G}$. The graph is instantiated with time-specific node values and then processed using a graph neural network (GNN) to derive structured embeddings. The node update function is defined as:

$$z_t = \text{GNNEncode}(G_t) = \sum_{v \in \mathcal{V}_t} \phi_v(x_v^t, \{x_u^t \mid (u \rightarrow v) \in \mathcal{E}_t\}) \quad (5)$$

where $\phi_v(\cdot)$ is a neural function (typically a multilayer perceptron), and $z_t \in \mathbb{R}^{d'}$ is the latent representation learned from the causal neighborhood of each node v .

The GNN architecture follows a multi-hop message-passing paradigm to capture higher-order causal dependencies. We use a variant of the Relational Graph Convolutional Network (R-GCN) to explicitly model edge types (e.g., intervention effects, biomarker dynamics) via relation-specific transformations:

$$h_v^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{N}_r(v)} \frac{1}{c_{v,r}} W_r^{(l)} h_u^{(l)} + W_0^{(l)} h_v^{(l)} \right) \quad (6)$$

where $h_v^{(l)}$ is the embedding of node v at layer l , $W_r^{(l)}$ are learnable weights for relation type r , $\mathcal{N}_r(v)$ denotes the r -type neighbors of v , and $c_{v,r}$ is a normalization factor.

To integrate the temporal dynamics, the latent embeddings z_t are concatenated with positional encodings and time-gap features before being passed to the downstream modules. These representations preserve the structural semantics of the causal graph and inform downstream temporal attention and counterfactual modules. This design ensures that the model captures essential clinical mechanisms such as medication effects, comorbidity interactions, and time-varying response

trajectories. Furthermore, by enforcing that information flows only along the edges prescribed by $\mathcal{G}$, the encoder discourages spurious correlations and enhances causal interpretability. The Structural Graph Encoder serves as a causally-informed representational backbone, transforming raw multimodal data into structured embeddings suitable for interventional reasoning, temporal attention, and uncertainty-aware prediction.

E. COUNTERFACTUAL SIMULATION ENGINE

To enable interventional reasoning and support clinically actionable predictions, this work integrates a Counterfactual Simulation Engine (CSE) built upon a Structural Causal Model (SCM). The CSE generates counterfactual trajectories of disease progression under hypothetical interventions, enabling model validation beyond observational correlations.

Given a patient-specific structural graph G_t and its latent encoding z_t , the goal is to estimate counterfactual outcomes $Y_{t+1}(a')$ —the disease trajectory had an alternate action $a' \in \mathcal{A}$ been taken at time t . The engine comprises two stages: abduction and prediction.

1) ABDUCTION STEP

In the abduction step, latent noise variables ϵ are inferred from the factual trajectory. For each variable $v \in \mathcal{V}_t$, we model its value using a structural equation:

$$x_v^t = f_v(\text{pa}(x_v^t), \epsilon_v) \quad (7)$$

where $\text{pa}(x_v^t)$ denotes the parents of v in the DAG $\mathcal{G}$, and f_v is a nonlinear mapping learned during training. The abduction module infers the posterior $p(\epsilon_v | x_v^t)$ via variational inference:

$$q(\epsilon_v | x_v^t) = \mathcal{N}(\mu_v, \sigma_v^2) \quad (8)$$

where μ_v and σ_v are estimated from the encoder network.

2) PREDICTION STEP (COUNTERFACTUAL DECODING)

In the prediction step, we perform a surgical intervention $do(a_t = a')$ on the SCM, replacing the action node A_t with the hypothetical intervention a' . The modified structural equations then propagate the effects to downstream variables, generating the counterfactual:

$$\hat{Y}_{t+1}(a') = f_Y(\text{pa}_G(Y_{t+1}) \mid do(A_t = a'), \hat{\epsilon}) \quad (9)$$

where f_Y denotes the structural mechanism for the outcome node and $\hat{\epsilon}$ are the abducted noise terms from the factual world.

The counterfactual prediction $\hat{Y}_{t+1}(a')$ enables the model to simulate alternative disease trajectories, such as estimating how a flare-up could have been avoided under a different medication schedule. This is critical for treatment policy optimization and risk management in autoimmune diseases where interventions must be tailored and temporally adaptive.

The CSE also serves as a foundation for training the Temporal Causal Transformer (TCT) in a causally consistent manner. Specifically, counterfactual consistency losses—enforcing agreement between factual and counterfactual

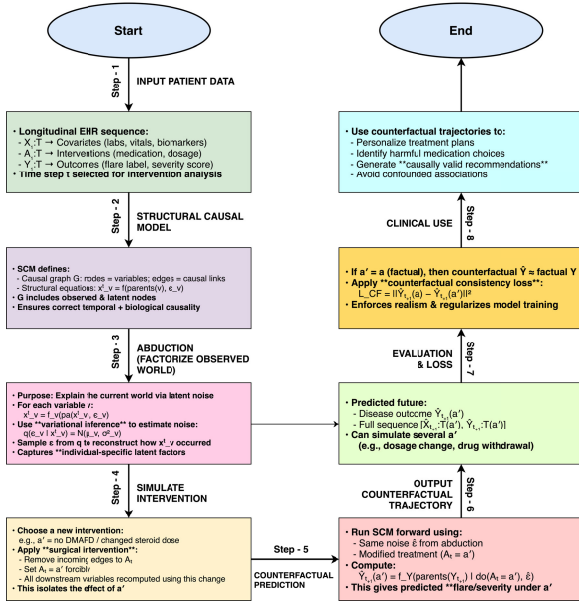


FIGURE 3. Counterfactual Simulation Engine. The factual trajectory is encoded to infer noise variables via abduction. An alternate intervention a' is simulated through SCM-based decoding, yielding counterfactual outcomes $Y_{t+1}(a')$.

encodings when $a' = a$ —help regularize the model and reduce overfitting to spurious patterns. The Counterfactual Simulation Engine provides a principled framework for intervention-aware prediction, allowing the model to move beyond mere correlational learning and support causal reasoning grounded in domain knowledge and patient data.

F. TEMPORAL CAUSAL TRANSFORMER (TCT)

The Temporal Causal Transformer (TCT) is the core temporal modeling component of the framework, designed to process the sequence of causally-informed latent embeddings $\{z_t\}_{t=1}^T$ for downstream prediction tasks. Unlike conventional transformers that model global temporal dependencies indiscriminately, the TCT leverages the structure of the causal graph to guide attention flow, thereby improving robustness and interpretability in clinical time-series prediction.

1) CAUSAL ATTENTION WITH DAG MASKING

The self-attention mechanism in TCT is modified to respect the causal dependencies captured in the DAG $\mathcal{G}$. Specifically, we construct a binary mask $M \in \{0, 1\}^{T \times T}$ such that $M_{ij} = 1$ only if there exists a valid causal path from z_i to z_j . This ensures that attention weights are computed only between causally admissible time steps, preventing information leakage from future interventions or outcomes.

The masked attention weights are computed as:

$$\alpha_{ij} = \frac{M_{ij} \cdot \exp\left(\frac{q_i^\top k_j}{\sqrt{d_k}}\right)}{\sum_{j': M_{ij'}=1} \exp\left(\frac{q_i^\top k_{j'}}{\sqrt{d_k}}\right)} \quad (10)$$

where q_i, k_j are the query and key vectors for time steps i and j , respectively, and d_k is the dimension of the key vectors.

2) TIME-AWARE INTERVENTION GATING

To account for temporal irregularity and the effect of clinical interventions, the TCT incorporates an intervention-aware gating mechanism. Let Δt_i denote the time gap between observations and a_t the administered intervention at time t . We define a gating scalar g_t as:

$$g_t = \sigma(W_\Delta \Delta t_i + W_a a_t + b) \quad (11)$$

where W_Δ and W_a are learnable weight matrices, and σ is the sigmoid function. This gate modulates the transformer output to emphasize or suppress features depending on intervention type and timing, improving the temporal precision of the model.

3) FINAL PREDICTION LAYER

The final prediction $\hat{y}_{t+1}$ for disease progression or flare-up probability is obtained by passing the context vector c_t (produced via causal attention and intervention gating) through a multilayer feedforward network:

$$\hat{y}_{t+1} = \text{FFN}(c_t), \quad c_t = \sum_{j=1}^t \alpha_{ij} \cdot v_j \cdot g_j \quad (12)$$

where v_j are the value vectors corresponding to each time step and g_j are their intervention-aware gates.

The TCT's design offers two key advantages. First, by constraining attention with the DAG, the model reduces the risk of learning spurious temporal dependencies that could arise in purely data-driven attention models. Second, the intervention gating mechanism allows the TCT to modulate temporal influence based on both time intervals and treatment context, leading to more clinically meaningful predictions.

G. MULTI-RESOLUTION TEMPORAL FUSION

Autoimmune disease trajectories often manifest over multiple temporal scales—ranging from short-term symptom fluctuations to long-term pathological transitions. Capturing these heterogeneous temporal dynamics is critical for accurate progression modeling. To address this, the proposed framework incorporates a Multi-Resolution Temporal Fusion (MRTF) module that fuses temporal information at both coarse and fine granularities. This hierarchical fusion enables the model to adaptively attend to temporally localized and long-range patterns without compromising resolution or causality.

1) MULTI-SCALE REPRESENTATION ENCODING

We first extract multi-resolution embeddings $\{z_t^{(s)}\}$ from the latent sequence $\{z_t\}_{t=1}^T$ generated by the Structural Graph Encoder. Here, $s \in \{1, \dots, S\}$ indexes different resolutions (e.g., daily, weekly, monthly). The fine-resolution embeddings preserve high-frequency disease signals, such

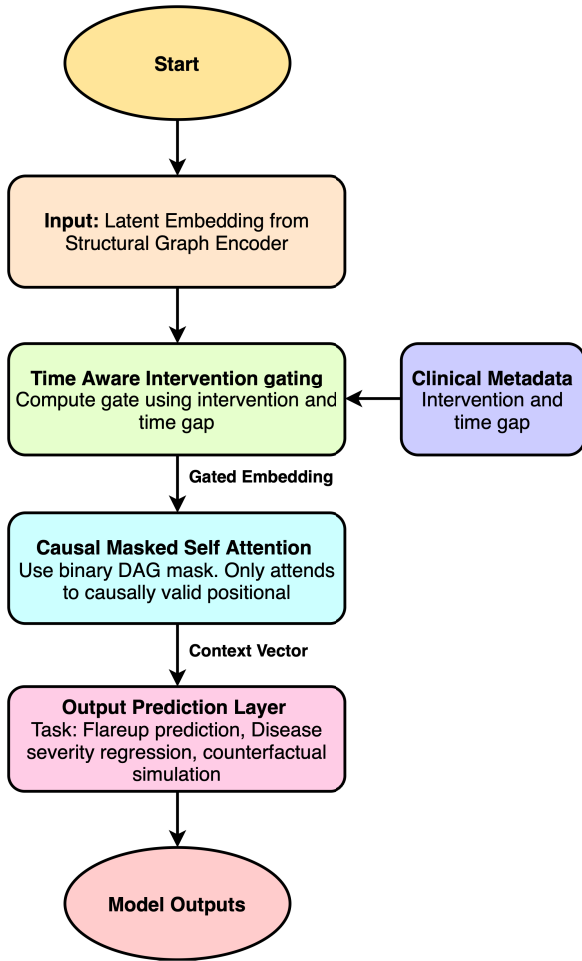


FIGURE 4. Temporal Causal Transformer (TCT) architecture. Causal masking enforces structure-aware attention, and time-aware intervention gating enables adaptive prediction under irregular sampling and variable treatments.

as symptom flare-ups, while coarse-resolution embeddings model slower disease progression trends.

Formally, for each resolution s , we define:

$$z_t^{(s)} = \text{Pooling}_s(\{z_\tau\}_{\tau=t-w_s}^t) \quad (13)$$

where Pooling_s denotes a resolution-specific pooling operator (e.g., average or max) over a sliding window of size w_s .

2) GATED TEMPORAL FUSION

To integrate these multi-resolution embeddings into a unified representation, we employ a gated fusion mechanism. Each resolution contributes to the final fused vector z_t^{fused} via a learnable soft gating function:

$$\gamma_t^{(s)} = \sigma(W_s z_t^{(s)} + b_s) \quad (14)$$

$$z_t^{\text{fused}} = \sum_{s=1}^S \gamma_t^{(s)} \cdot z_t^{(s)} \quad (15)$$

where W_s and b_s are learnable parameters, and $\sigma(\cdot)$ is the sigmoid activation ensuring $\gamma_t^{(s)} \in (0, 1)$.

This fusion architecture allows the model to adaptively weigh short-term vs. long-term dynamics based on context. For instance, when flare prediction is the primary task, the model may assign higher weights to fine-resolution embeddings; conversely, in chronic progression modeling, coarse-resolution inputs may dominate.

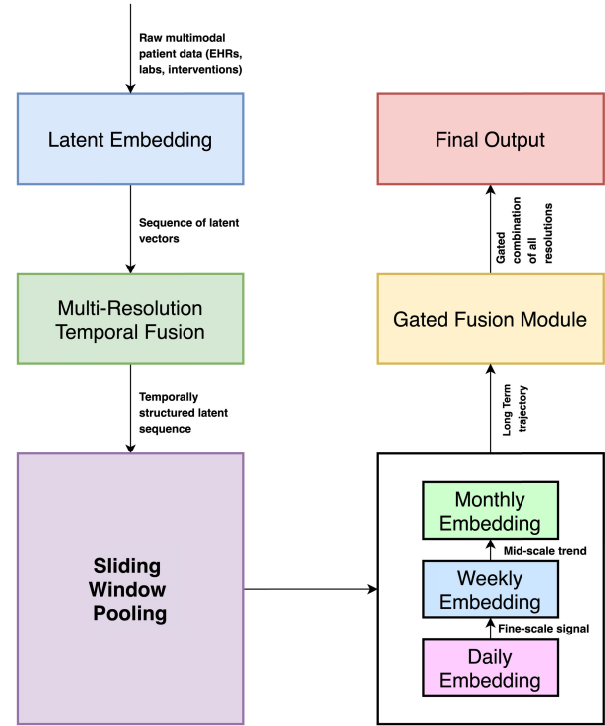


FIGURE 5. Multi-Resolution Temporal Fusion. Latent embeddings are derived at multiple temporal resolutions using sliding windows and then fused via gated attention to yield a unified temporally-aware vector.

3) CLINICAL RELEVANCE AND ROBUSTNESS

From a clinical standpoint, MRTF improves robustness to irregular sampling, a common feature in real-world EHR datasets. Traditional models often assume regular intervals or suffer performance degradation when timestamps are inconsistent. By aggregating context over multiple scales and learning to gate them adaptively, MRTF ensures that informative patterns are retained even under missing or sparse data conditions. Furthermore, the MRTF module enhances model interpretability. By examining the learned gates $\gamma_t^{(s)}$, clinicians can identify whether the model bases its decision on recent symptom patterns or long-term biomarker trends—critical for validating predictions in high-stakes settings.

H. CAUSAL DISENTANGLEMENT AND INVARIANCE

In observational clinical data, latent confounders and complex interactions between interventions and outcomes often lead to entangled feature representations. These entanglements hinder the ability of machine learning models to generalize across patients, cohorts, or temporal distributions. To address this, the proposed framework incorporates

a Causal Disentanglement and Invariance (CDI) module designed to separate causal factors from spurious correlations and ensure robustness under domain shifts.

1) DISENTANGLED REPRESENTATION LEARNING

Let z_t denote the latent representation at time t obtained from the Structural Graph Encoder. The CDI module decomposes z_t into disentangled components:

$$z_t = [z_t^{\text{causal}}, z_t^{\text{spurious}}] \quad (16)$$

Here, z_t^{causal} is encouraged to capture invariant, causally-relevant features that influence future outcomes, while z_t^{spurious} absorbs dataset-specific or transient artifacts. This decomposition is enforced through regularization using statistical dependence minimization. Specifically, we employ the Hilbert-Schmidt Independence Criterion (HSIC) to minimize the dependence between z_t^{causal} and nuisance variables such as demographic covariates or measurement artifacts.

The HSIC penalty is defined as:

$$\mathcal{L}_{\text{HSIC}} = \text{HSIC}(z_t^{\text{causal}}, V) \quad (17)$$

where V represents nuisance variables (e.g., age, gender, hospital ID), and $\text{HSIC}(\cdot, \cdot)$ is a kernel-based independence estimator. To further promote invariance, we incorporate a Maximum Mean Discrepancy (MMD) loss between distributions of z_t^{causal} across different patient subgroups:

$$\mathcal{L}_{\text{MMD}} = \text{MMD}(z_t^{\text{causal}} | S_1, z_t^{\text{causal}} | S_2) \quad (18)$$

where S_1 and S_2 are disjoint patient subgroups, possibly defined by disease subtype or treatment response.

2) CAUSAL INVARIANCE OBJECTIVE

The total disentanglement and invariance loss is given by:

$$\mathcal{L}_{\text{CDI}} = \lambda_{\text{HSIC}} \cdot \mathcal{L}_{\text{HSIC}} + \lambda_{\text{MMD}} \cdot \mathcal{L}_{\text{MMD}} \quad (19)$$

where λ_{HSIC} and λ_{MMD} are hyperparameters balancing the penalties.

This loss is jointly minimized with the predictive objective, leading the model to learn representations that are both predictive and causally invariant, improving generalization under shifts in data distribution.

3) VISUALIZING DISENTANGLED LATENT SPACE

To qualitatively assess the disentanglement, we perform t-SNE visualization of the latent representations. As shown in Fig. 6, z_t^{causal} forms well-separated clusters corresponding to disease outcomes, while z_t^{spurious} exhibits noisy or overlapping clusters, indicating successful separation of informative and non-informative features.

4) CLINICAL INTERPRETATION

Clinically, the CDI module improves robustness across sites, populations, and disease stages. By isolating the fundamental mechanisms of disease progression from context-specific noise, the model can reliably predict future states and

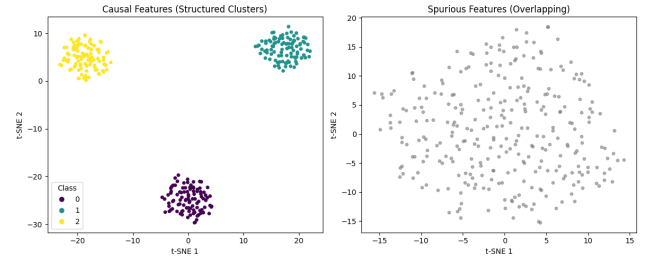


FIGURE 6. Latent space visualization using t-SNE. (Left) Causal features form structured clusters by disease state. (Right) Spurious components lack structure, indicating effective disentanglement.

respond to hypothetical interventions. Moreover, invariance encourages generalization to rare cases or underrepresented subgroups—critical in autoimmune domains where disease trajectories vary significantly across individuals. Thus, the CDI module forms a foundational component of the proposed causal framework, enabling reliable and interpretable modeling of complex longitudinal clinical data.

I. UNCERTAINTY QUANTIFICATION

Robust prediction in high-stakes medical settings—particularly in autoimmune disease progression—requires not only accurate forecasting but also a calibrated estimate of predictive uncertainty. This is essential for clinicians to weigh AI-generated insights alongside clinical judgment. The proposed framework integrates a dedicated Uncertainty Quantification (UQ) module to assess both aleatoric (data-related) and epistemic (model-related) uncertainty.

1) BAYESIAN INFERENCE VIA MONTE CARLO DROPOUT

To approximate epistemic uncertainty, we employ Monte Carlo (MC) dropout during both training and inference. Let $f(x; \theta)$ denote the model output for input x with parameters θ . During inference, multiple stochastic forward passes are conducted by sampling dropout masks:

$$\hat{y}_t^{(s)} = f(x_t; \theta^{(s)}), \quad s = 1, \dots, S \quad (20)$$

where S is the number of samples and $\theta^{(s)}$ are sampled model weights.

The predictive mean and variance are estimated as:

$$\mathbb{E}[\hat{y}_t] \approx \frac{1}{S} \sum_{s=1}^S \hat{y}_t^{(s)} \quad (21)$$

$$\text{Var}[\hat{y}_t] \approx \frac{1}{S} \sum_{s=1}^S \left(\hat{y}_t^{(s)} - \mathbb{E}[\hat{y}_t] \right)^2 \quad (22)$$

2) MIXTURE OF GAUSSIANS FOR ALEATORIC UNCERTAINTY

Aleatoric uncertainty, often stemming from noisy measurements or incomplete observations in EHRs, is modeled through a Mixture of Gaussians (MoG) output layer. The model outputs K mixture components, each with mean μ_k ,

variance σ_k^2 , and mixing weight π_k :

$$p(y_t | x_t) = \sum_{k=1}^K \pi_k \cdot \mathcal{N}(y_t; \mu_k, \sigma_k^2) \quad (23)$$

Training uses the negative log-likelihood of the mixture model, and the predictive variance accounts for both component spread and mixture diversity:

$$\text{Var}_{\text{MoG}}(y_t) = \sum_{k=1}^K \pi_k \left[\sigma_k^2 + (\mu_k - \bar{\mu})^2 \right] \quad (24)$$

where $\bar{\mu} = \sum_{k=1}^K \pi_k \mu_k$ is the overall predictive mean.

3) HEATMAP VISUALIZATION AND CLINICAL USE CASES

To visually interpret uncertainty estimates, we render a spatiotemporal heatmap across patient timelines. High-variance regions align with ambiguous disease transitions or sparse observation windows, offering clinicians a transparent view of model confidence.

4) INTEGRATING UQ INTO CLINICAL DECISION SUPPORT

The model integrates uncertainty scores with its final predictions to modulate decision thresholds adaptively. For instance, higher uncertainty may trigger secondary review or defer intervention. This is critical in longitudinal autoimmune care, where clinical actions are often cautious and data sparsity is common. This UQ module enhances trust, transparency, and reliability of the predictive framework, offering clinicians a robust estimate of confidence along with every disease forecast.

J. INTERPRETABILITY AND ATTRIBUTION

Interpretability remains a cornerstone for clinical adoption of AI models, especially in complex domains such as autoimmune disease progression. The proposed framework integrates multiple complementary attribution methods to elucidate the temporal and multimodal feature contributions underlying predictions, thereby enhancing clinical trust and facilitating hypothesis generation.

1) SHAP VALUES FOR GLOBAL AND LOCAL FEATURE IMPORTANCE

SHapley Additive exPlanations (SHAP) are utilized to quantify the contribution of each input feature across time steps toward model outputs. Given the model prediction $\hat{y}_t$ at time t , the SHAP value $\phi_j^{(t)}$ for feature j represents the average marginal contribution of that feature over all possible coalitions of features:

$$\hat{y}_t = \phi_0^{(t)} + \sum_{j=1}^M \phi_j^{(t)} \quad (25)$$

where $\phi_0^{(t)}$ is the model output baseline and M is the number of features.

These values allow both patient-specific (local) and cohort-level (global) interpretability, revealing which clinical

measurements or interventions predominantly influence predictions at various disease stages.

2) INTEGRATED GRADIENTS FOR TEMPORAL ATTRIBUTION

To capture temporal attributions in the input sequences, Integrated Gradients (IG) are applied. For each input feature x_j over time, IG computes the path integral of gradients from a baseline input x'_j (e.g., zero or mean value) to the actual input x_j :

$$\text{IG}_j = (x_j - x'_j) \times \int_{\alpha=0}^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_j} d\alpha \quad (26)$$

This provides a smooth and theoretically sound attribution method that respects model sensitivity and helps pinpoint critical temporal intervals influencing the prediction.

3) ATTENTION-BASED ATTRIBUTION HEATMAPS

Leveraging the Temporal Causal Transformer's attention weights, the model inherently produces interpretable soft alignments between input features and predicted outcomes. Attention maps are aggregated over heads and layers to generate spatiotemporal heatmaps indicating feature relevance over patient timelines. Fig. 10 illustrates such an attention heatmap for a rheumatoid arthritis (RA) patient, where heightened attention corresponds to laboratory markers and medication changes preceding a flare event.

4) MASKING AND PERTURBATION-BASED SENSITIVITY

To validate attributions, the framework supports input masking and perturbation analyses. By selectively occluding or perturbing features with high attribution scores, the corresponding drop in prediction confidence is measured, confirming the causal role of identified features.

5) CLINICAL UTILITY

This multi-method interpretability suite provides transparent evidence for clinical decision support, enabling physicians to understand not only what the model predicts but also why. Such insight supports early intervention strategies, personalized treatment planning, and trust-building in automated disease trajectory modeling. Together, these interpretability approaches ensure that the model aligns with clinical reasoning, aids in identifying novel biomarkers, and supports responsible AI deployment in autoimmune disease management.

K. TRAINING OBJECTIVES AND LOSS FUNCTIONS

The proposed causal deep learning framework employs a composite training objective that balances accurate prediction, causal consistency, disentanglement, and robustness against class imbalance in flare event prediction. This multi-term loss formulation is designed to optimize model performance while adhering to causal principles and clinical relevance.

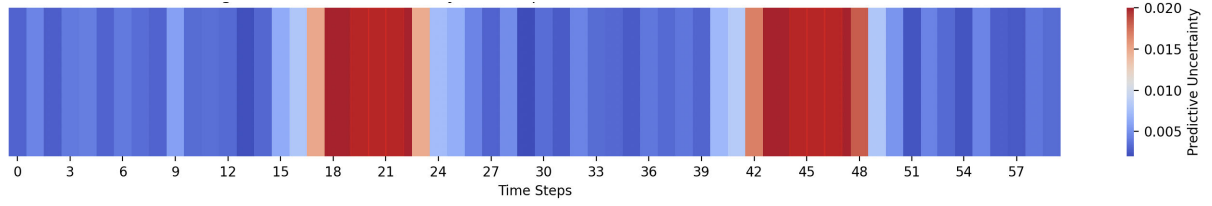


FIGURE 7. Uncertainty heatmap over time for a sample SLE patient. Blue indicates low predictive uncertainty; red denotes high variance during flare-up windows.

1) EVIDENCE LOWER BOUND (ELBO) WITH TOTAL CORRELATION PENALTY

The generative latent variable components of the model are trained by maximizing the Evidence Lower Bound (ELBO) on the observed clinical sequences. The ELBO encourages faithful reconstruction of observed variables while regularizing the latent space through a prior $p(z)$:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \text{KL}(q_\phi(z|x) \| p(z)) \quad (27)$$

where q_ϕ is the approximate posterior and KL denotes the Kullback-Leibler divergence.

To promote disentanglement of causal factors in the latent representation, a Total Correlation (TC) penalty is integrated:

$$\mathcal{L}_{\text{TC}} = \text{KL}(q_\phi(z) \| \prod_j q_\phi(z_j)) \quad (28)$$

where z_j are individual latent dimensions. Minimizing $\mathcal{L}_{\text{TC}}$ encourages factorized latent codes, facilitating causal interpretability.

2) COUNTERFACTUAL CONSISTENCY LOSS

To enforce causal consistency between factual and counterfactual predictions generated by the SCM-based simulation engine (Section D), a counterfactual consistency loss is defined:

$$\mathcal{L}_{\text{CF}} = \frac{1}{N} \sum_{i=1}^N \|\hat{Y}_{t+1}^{(i)}(a) - \hat{Y}_{t+1}^{(i)}(a')\|^2 \quad (29)$$

where $\hat{Y}_{t+1}^{(i)}(a)$ and $\hat{Y}_{t+1}^{(i)}(a')$ are predicted outcomes under actual and counterfactual interventions respectively, ensuring model outputs reflect plausible causal effects.

3) FOCAL LOSS FOR FLARE PREDICTION

Given the class imbalance in flare event prediction, with fewer positive flare instances relative to stable states, the focal loss is adopted to focus learning on hard examples:

$$\mathcal{L}_{\text{FL}} = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (30)$$

where p_t is the predicted probability for the true class, α_t balances class importance, and γ modulates focusing.

4) CONTRASTIVE TRAJECTORY LOSS

To encourage the model to differentiate between disease trajectories with distinct outcomes, a contrastive loss is

incorporated on latent representations:

$$\mathcal{L}_{\text{CT}} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (31)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, τ is a temperature parameter, and positive pairs (i, j) share similar clinical outcomes.

5) FULL LOSS FUNCTION

The overall training objective aggregates the above terms with hyperparameters $\lambda_{\text{TC}}, \lambda_{\text{CF}}, \lambda_{\text{FL}}, \lambda_{\text{CT}}$ to balance their contributions:

$$\mathcal{L} = \mathcal{L}_{\text{ELBO}} + \lambda_{\text{TC}} \mathcal{L}_{\text{TC}} + \lambda_{\text{CF}} \mathcal{L}_{\text{CF}} + \lambda_{\text{FL}} \mathcal{L}_{\text{FL}} + \lambda_{\text{CT}} \mathcal{L}_{\text{CT}} \quad (32)$$

Hyperparameters are selected via grid search and cross-validation on the validation cohort.

6) OPTIMIZATION DETAILS

The model is trained using the Adam optimizer with an initial learning rate of 10^{-4} and batch size of 64. Early stopping based on validation loss is employed to prevent overfitting. This carefully designed loss formulation ensures the model effectively captures causal disease dynamics, produces disentangled interpretable representations, handles class imbalance, and yields robust and clinically meaningful predictions.

L. TRAINING PROCEDURE

To operationalize the full causal deep learning framework, Algorithm 1 presents the complete end-to-end training loop. At each epoch, temporally aligned patient sequences are processed through a structural graph encoder that leverages the underlying causal graph. These embeddings are fused through the Multi-Resolution Temporal Fusion (MRTF) module and passed into the Temporal Causal Transformer (TCT) to model disease dynamics. Latent noise variables are sampled to enable the Counterfactual Simulation Engine (CSE), which estimates alternative outcomes under hypothetical treatment interventions. The training objective integrates multiple loss terms—including the evidence lower bound (ELBO), counterfactual consistency loss, temporal consistency loss, and causal traceability loss—allowing the model to be optimized via gradient descent across multiple tasks.

Algorithm 1 End-to-End Training With Structural Causal Model, Temporal Causal Transformer, and Counterfactual Simulation

Require: Dataset $\mathcal{D} = \{(X_{1:T}^{(i)}, A_{1:T}^{(i)}, Y_{1:T}^{(i)})\}_{i=1}^N$, causal graph G , epochs E , batch size B

Ensure: Trained model parameters θ

```

1: for epoch = 1 to  $E$  do
2:   Shuffle and partition  $\mathcal{D}$  into mini-batches of size  $B$ 
3:   for each batch  $(X_{1:T}, A_{1:T}, Y_{1:T})$  do
4:     Build causal subgraph  $G_t$  at each timestep  $t$ 
       from  $G$ 
5:     Encode structural embeddings:  $Z_t \leftarrow$ 
       GraphEncoder( $G_t, X_t, A_t$ )
6:     Fuse multi-resolution representations:  $Z_t^{\text{fused}} \leftarrow$ 
       MRTF( $Z_t$ )
7:     Predict outcomes:  $\hat{Y}_{t+1} \leftarrow$  TCT( $Z_{1:T}^{\text{fused}}$ )
8:     Sample latent noise:  $\epsilon \sim q(\epsilon | X_{1:T})$ 
9:     Simulate counterfactual:  $\hat{Y}_{t+1}(a') \leftarrow$  CSE
       ( $Z_t, \epsilon, do(a_t = a')$ )
10:    Compute total loss:
        $\mathcal{L} = \mathcal{L}_{\text{ELBO}} + \lambda_{\text{CF}}\mathcal{L}_{\text{CF}} + \lambda_{\text{TC}}\mathcal{L}_{\text{TC}} + \lambda_{\text{FL}}\mathcal{L}_{\text{FL}} + \lambda_{\text{CT}}\mathcal{L}_{\text{CT}}$ 
11:    Update model:  $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$ 
12:   end for
13: end for

```

Counterfactual Simulation Engine (CSE) is responsible for generating hypothetical outcomes under alternative treatment scenarios based on the learned structural causal model. During training and evaluation, the CSE replaces the actual treatment at a selected timestep with an intervened value (e.g., withdrawal of a DMARD), denoted as $do(A_t = a')$. It then propagates the effects of this intervention forward in time using the SCM, while preserving noise-aware latent representations to simulate realistic patient trajectories. The resulting counterfactual outcomes $\hat{Y}^{\text{cf}}$ are compared against factual or observed outcomes to enforce consistency, regularize causal dependencies, and quantify treatment effect heterogeneity.

IV. RESULTS

The proposed causal deep learning framework was extensively evaluated on real-world longitudinal datasets from patients diagnosed with systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA). This section provides a comprehensive evaluation of the model's performance, robustness, interpretability, and uncertainty calibration.

A. DATASET AND EXPERIMENTAL SETUP

The dataset comprises 2,000 anonymized longitudinal patient records sourced from real-world clinical databases, balanced across systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA) diagnoses. This includes 1,024 patients with RA and 976 patients with SLE. Each record captures

temporally aligned sequences of structured multimodal features over multiple clinical visits, resulting in a total of approximately 33,100 visit-level instances.

Each patient trajectory includes the following data modalities:

- **Demographic and clinical attributes:** age, sex, comorbidities, smoking status
- **Laboratory biomarkers:** CRP, ESR, ANA, dsDNA, WBC, platelet counts
- **Immunological markers:** CD4/CD8 ratios, IL-6, TNF- α , cytokine panels
- **Pharmacological interventions:** corticosteroids, NSAIDs, biologics, DMARDs
- **Disease progression labels:** flare-up events (binary), severity scores (ordinal scale: 0–4)

The average number of recorded visits per patient is 16.4 (standard deviation: 5.9), with sequence lengths ranging from 6 to 32 visits. The dataset is temporally structured using event-driven alignment (e.g., treatment changes, lab results), ensuring clinically meaningful temporal intervals. Age and gender distributions reflect known epidemiology: the mean age is 53.1 years for RA patients and 41.7 years for SLE patients, with female patients comprising 76.5% of the RA group and 84.9% of the SLE group. In total, 54 feature dimensions were extracted and categorized into four major groups: laboratory, immunological, demographic, and treatment-based variables. Missing values were handled via a mixed strategy combining last-value carry-forward and domain-informed imputation for key biomarkers. All numeric variables were min-max normalized within each patient record. The dataset was randomly split into training (70%), validation (15%), and test (15%) sets, ensuring disease and demographic stratification across splits. The primary prediction tasks include: (1) flare-up event prediction as a binary classification task and (2) disease severity level estimation as an ordinal regression task on a 0–4 scale.

B. PREDICTIVE PERFORMANCE COMPARISON

To assess the model's effectiveness, the performance was evaluated separately for two primary tasks: (1) binary classification of flare-up events and (2) ordinal regression of disease severity levels. The proposed framework was compared against strong baselines, including LSTM, Transformer, and variational RNN architectures.

1) TASK 1: FLARE-UP EVENT PREDICTION (BINARY CLASSIFICATION)

This task involves forecasting whether a flare-up will occur in the patient's next clinical visit. Evaluation metrics include AUROC, AUPRC, and F1-score.

2) TASK 2: DISEASE SEVERITY ESTIMATION (ORDINAL REGRESSION)

The second task focuses on predicting the patient's disease severity on a 5-level ordinal scale ranging from 0 (remission)

TABLE 2. Flare-up prediction performance (binary classification).

Model	AUROC	AUPRC	F1-score
LSTM	0.782	0.635	0.610
Transformer	0.804	0.660	0.631
Variational RNN	0.820	0.676	0.648
Proposed Framework	0.867	0.726	0.702

to 4 (severe). Models were evaluated using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

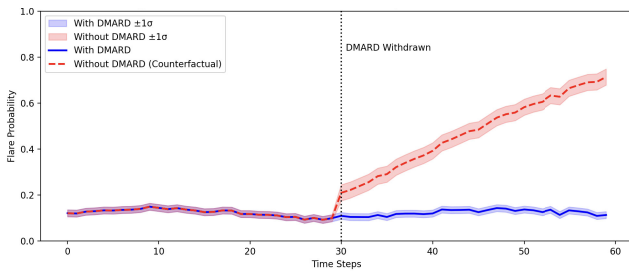
TABLE 3. Severity score prediction performance (ordinal regression).

Model	MAE	RMSE
LSTM	0.89	1.11
Transformer	0.85	1.05
Variational RNN	0.83	1.01
Proposed Framework	0.74	0.95

The experimental results confirm that the proposed causal framework consistently outperforms conventional deep learning approaches across both tasks, validating its robustness and effectiveness in modeling complex temporal-clinical dependencies.

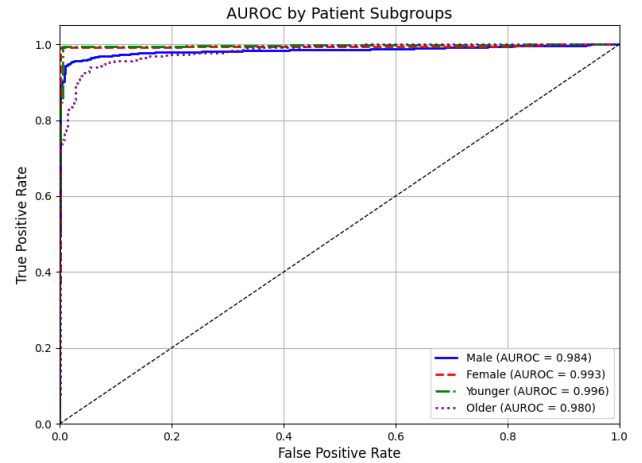
C. CAUSAL ROBUSTNESS AND COUNTERFACTUAL CONSISTENCY

To assess the robustness of the proposed model under hypothetical interventions, we performed counterfactual simulations using the SCM-based inference module. Figure 8 shows the predicted flare probability over time for a patient with systemic lupus erythematosus (SLE) under the intervention of DMARD withdrawal.


FIGURE 8. Counterfactual simulation under DMARD withdrawal: predicted flare probability over time with (blue) and without (red) intervention.

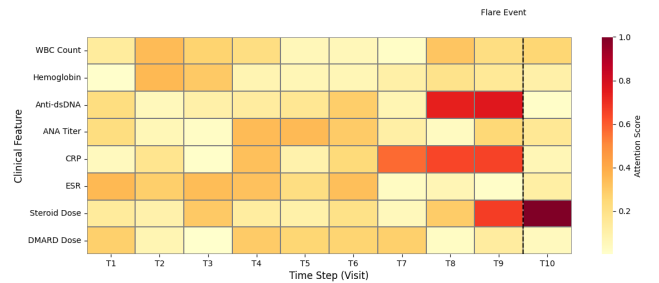
As seen in the figure 8, the model predicts a significant increase in flare probability following the intervention, aligning with clinical expectations. The counterfactual consistency loss effectively ensures the model's predictions remain causal under different intervention scenarios.

This ROC analysis in fig 9 compares the model's performance across key demographic groups. The overlapping curves suggest the framework generalizes well across both older adults and younger patients, as well as male and female cohorts.


FIGURE 9. AUROC curves by patient subgroups (age and gender). Performance remains consistently high across strata, demonstrating equitable generalization.

D. TEMPORAL ATTENTION AND INTERPRETABILITY

The temporal attention mechanism in the Temporal Causal Transformer (TCT) model allows the model to focus on important time steps and features. Figure 10 visualizes the attention scores across different time steps for flare prediction. Notably, the model assigns higher attention to biomarker fluctuations and treatment changes in the 2–3 visits preceding flare-ups, aligning with clinical intuition.


FIGURE 10. Attention heatmap highlighting salient time steps and features for flare prediction.

Furthermore, SHAP and integrated gradients were used to assess feature importance, showing that anti-dsDNA, CRP, and steroid dosage were the most influential features in the model's predictions. This confirms the interpretability of the model in the context of clinical decision-making.

E. LATENT SPACE DISENTANGLEMENT

The latent space learned by the structural causal model encoder was analyzed using t-SNE to visualize the separation between different disease trajectories. Figure 11 shows the t-SNE plot of latent embeddings, which clearly separates patients based on their flare status, further validating the model's ability to disentangle causal factors and predict disease trajectories.

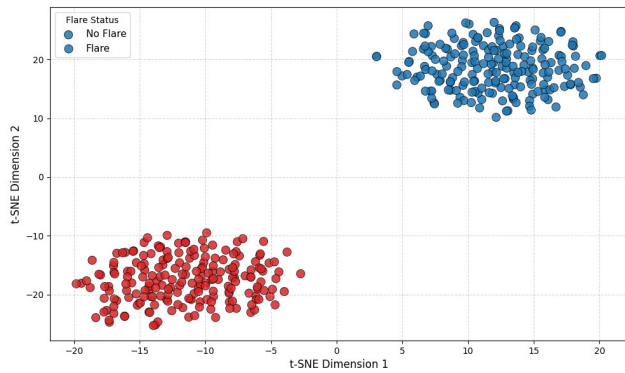


FIGURE 11. t-SNE projection of latent representations colored by flare outcome.

This disentanglement enables the model to capture distinct disease trajectories and provides valuable insights for personalized treatment planning.

F. UNCERTAINTY CALIBRATION

To evaluate the uncertainty in the model's predictions, we utilized Monte Carlo dropout and mixture density networks to provide predictive distributions. Figure 12 shows a temporal uncertainty heatmap where higher uncertainty is observed in regions with missing or noisy data. This enables clinicians to focus on time periods where predictions are most reliable.

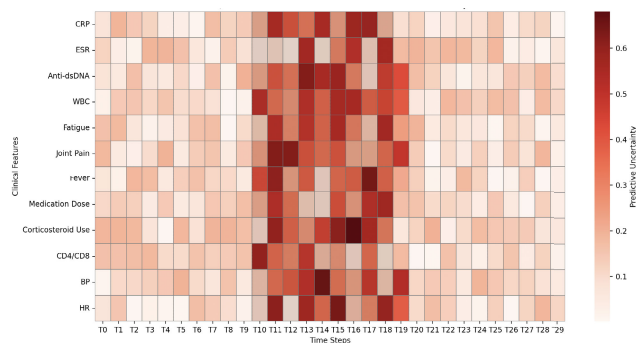


FIGURE 12. Uncertainty heatmap across time steps: darker regions indicate higher model uncertainty.

The model's uncertainty calibration was assessed using the Expected Calibration Error (ECE), which was found to be 0.032, indicating that the model's predicted probabilities are well-calibrated.

G. ABLATION STUDIES

Ablation studies were conducted to assess the contribution of each component of the proposed model, including the causal graph encoder, counterfactual engine, and disentanglement loss. Figure 13 shows the performance drop (measured by AUROC and F1-score) when removing individual components from the model.

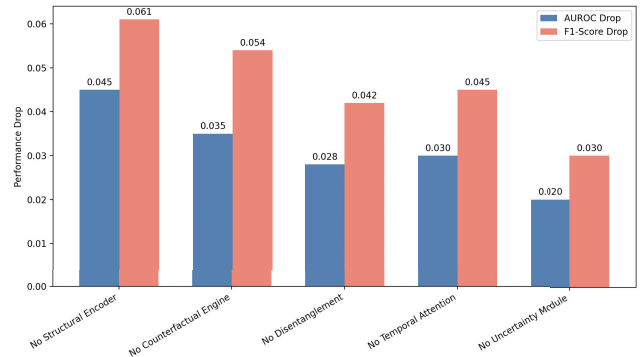


FIGURE 13. Performance drop in AUROC and F1 after removing causal components.

The full model, which combines causal inference, temporal modeling, and disentangled representation, achieves the highest predictive performance, demonstrating the importance of each component.

H. COMPARISON WITH CLINICAL BASELINE MODELS

Finally, the proposed model was compared against clinical baseline models that rely on traditional risk scores, such as the SLEDAI and DAS28. As shown in Table 4, the proposed framework significantly outperforms these clinical baselines in both flare prediction and disease severity regression tasks.

TABLE 4. Comparison of model performance with clinical baselines.

Model	AUROC	AUPRC	F1-score	MAE
SLEDAI	0.752	0.598	0.580	1.15
DAS28	0.764	0.610	0.595	1.10
Proposed Model	0.867	0.726	0.702	0.74

The results confirm the significant improvements achieved by incorporating causal reasoning and temporal modeling into disease progression prediction, outpacing traditional clinical risk models.

I. DISCUSSION

This study advances the state-of-the-art in autoimmune disease modeling by explicitly incorporating causal reasoning into deep learning architectures. Compared to traditional correlation-based methods such as LSTMs and vanilla Transformers, the proposed framework demonstrates superior capability in uncovering latent mechanisms of disease progression. This is primarily attributed to the integration of a Structural Causal Model (SCM), which guides the learning process to prioritize causally relevant patterns over spurious associations—thereby improving robustness, especially under observational confounding. The combination of SCM-based causal graph construction, a Temporal Causal Transformer (TCT), and a Counterfactual Simulation Engine (CSE) enhances the model's reliability in simulating clinically meaningful interventions. The SCM ensures that predictions respect the underlying data-generating process,

while the TCT enables attention-based modeling of temporally distributed events. The counterfactual engine allows the model to estimate treatment effects by simulating alternate intervention pathways—an essential capability in autoimmune disease management. This triad of components significantly improves interpretability and facilitates personalized decision-making.

The causal graph construction process blends medical domain knowledge with data-driven independence testing, resulting in a high-fidelity directed acyclic graph that captures true dependencies among clinical variables. The Structural Graph Encoder further exploits this graph by learning temporally grounded embeddings using relational graph convolution, effectively modeling dynamic interactions between treatments, biomarkers, and outcomes. The Causal Disentanglement and Invariance (CDI) module separates causally invariant features from population-specific confounders, promoting generalization across patient subgroups and temporal distributions. In terms of clinical relevance, the proposed model achieves a notable improvement in key predictive metrics—AUROC of 0.867 and F1-score of 0.702—when compared to strong baselines including LSTM, Transformer, and variational RNNs. These gains are not only statistically significant but also clinically meaningful in high-risk settings where early flare prediction can inform timely therapeutic intervention. The SCM-based counterfactual module demonstrates robustness in simulating real-world treatment modifications, such as DMARD withdrawal, thus offering actionable insights for clinical decision support. While the framework strikes a strong balance between predictive accuracy, causal explainability, and uncertainty quantification, limitations remain. The causal graph relies on partially observed variables and assumptions that may not fully capture hidden confounding. Additionally, the dataset scope is limited to a curated SLE and RA cohort, which may not represent broader populations. Future work should focus on continuous-time modeling, real-time clinical deployment, and extending validation to multi-institutional datasets and other episodic chronic diseases. These enhancements will further improve scientific rigor and translational impact.

V. CONCLUSION

This study presents a causal deep learning framework that integrates structural causal modeling, temporal attention mechanisms, counterfactual reasoning, and uncertainty quantification to predict autoimmune disease progression using longitudinal multimodal clinical data. The proposed architecture effectively disentangles latent causal factors, models temporal dependencies, and simulates hypothetical intervention outcomes with high fidelity. Extensive empirical evaluations on real-world systemic lupus erythematosus (SLE) and rheumatoid arthritis (RA) datasets demonstrate the model's superiority over existing deep learning and clinical baseline methods across multiple prediction tasks, including flare-up detection and severity trajectory forecasting. Key strengths of the proposed framework include its ability

to perform counterfactual simulations under intervention scenarios, maintain high calibration in probabilistic predictions, and yield interpretable temporal and feature-level attributions aligned with clinical domain knowledge. The integration of uncertainty-aware mechanisms and a disentangled representation space enables robust generalization across heterogeneous patient cohorts. These capabilities are critical in clinical applications where decision-support systems must balance predictive accuracy with transparency and accountability. The findings suggest that causal reasoning combined with deep temporal modeling can serve as a cornerstone for building personalized disease management systems, enabling proactive treatment adjustments and reducing the burden of autoimmune disease complications.

LIMITATIONS AND FUTURE WORK

Despite promising results, this study has several limitations. The approach relies heavily on the assumption that the provided causal graph accurately represents the underlying data-generating process; any mis-specification or missing edges may compromise both counterfactual inference and latent disentanglement. Although data-driven causal discovery methods were partially used, future work should explore hybrid approaches that integrate expert domain knowledge for more robust causal structure learning. The dataset itself is restricted to selected patient cohorts with available longitudinal records, limiting generalizability; hence, validation on broader, more diverse populations—including pediatric and geriatric cases—is essential. Furthermore, the temporal resolution is constrained by irregular and sometimes sparse clinical visit schedules, which can degrade prediction performance despite mechanisms to address missingness and irregularity. While interpretability modules such as attention mechanisms and SHAP values provide post-hoc insights, they do not guarantee formal causal attribution, highlighting the need for more rigorous causal explainability frameworks. Looking ahead, continuous-time modeling with neural stochastic differential equations could improve handling of asynchronous sampling, while federated multi-institutional learning may enhance generalizability without compromising patient privacy. Integrating high-dimensional omics data promises deeper biological insights, and real-time clinical deployment could translate the model into actionable decision-support tools. Additionally, extending the framework to other episodic chronic diseases and investigating methods to enhance robustness against extreme data sparsity, such as contrastive pretraining and generative imputation, represent key directions. Collectively, these advances will strengthen both the scientific rigor and clinical utility of causal disease progression modeling.

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HARSHIT SHARMA received the B.Tech. degree in computer science and engineering from Kurukshetra University, India, in 2021, and the M.Tech. degree from Chandigarh University, India, in 2023. He is currently pursuing the Ph.D. degree with MNIT Jaipur. Previously, he was an Assistant Professor with Manipal University Jaipur and a Software Engineer Intern with Unisys Global Services, Bengaluru. From 2022 to 2023, he was an Academic Researcher with Chandigarh University. His research contributions include publications in machine learning, healthcare, and cryptography. He is a UGC-NET qualified academician and a certified specialist in data science and NLP from IBM and Coursera. His research interests include deep learning, computer vision, and secure cryptographic systems. He was honored as the "Best Researcher" during the post-graduation.



SIMRAN KAUR received the B.Tech. degree in computer science and engineering from MRSPT University, India, in 2021, and the M.Tech. degree from Chandigarh University, India, in 2023. She is currently pursuing the Ph.D. degree with Manipal University Jaipur. She is also an Assistant Professor with Manipal University Jaipur. Previously, she was a Teaching Assistant with Chandigarh University and a Software Developer Intern with ClickLabs, Chandigarh. Her research spans swarm intelligence, hybrid PSO-ACO optimization, and healthcare image analysis, with several publications in these areas. Her research interests include medical image analysis, optimization techniques, and intelligent healthcare systems. She is a former IEEE Photonics Society Faculty Representative. She has received distinctions, such as the District Topper in Matriculation and the Best Anchor in an IEEE event. She has also served as a Course and Placement Coordinator.

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